

Smart Attendance Verification System

RFID Check-In + BLE Dwell-Time Tracking (40+ Min Verification Logic)

PROJECT TYPE	TARGET DEPLOYMENT	ESTIMATED PROTOTYPE COST
Embedded IoT & Automation	Smart Classrooms & Lecture Halls	₹1,350 – ₹1,950 (~\$16 – \$23)

1. EXECUTIVE SUMMARY & SYSTEM CONCEPT

Traditional RFID attendance systems suffer from a major vulnerability: **"Proxy Tapping"** or "Proxy Marking", where a student taps their card at the door and immediately leaves the room.

This proposed **Smart Attendance Verification System** solves proxy tapping by implementing a two-layer verification state machine combining explicit physical check-in with continuous passive presence monitoring:

- Layer 1 (Entry Check-in):** Student taps their RFID ID card at the door RC522 reader upon entering. This initializes an active session in the microcontroller buffer.
- Layer 2 (BLE Dwell-Time Verification):** An ESP32 microcontroller continuously scans for the student's unique BLE beacon/smartphone signal inside the classroom. If the RSSI signal strength indicates presence inside the room (e.g., > -75 dBm) for a cumulative duration of at least **40 minutes** during a 50–60 minute session, the attendance state transitions from PENDING to VERIFIED_PRESENT and gets logged to the database via Wi-Fi/MQTT.

Key Innovation

If a student taps their RFID card and leaves, the BLE scan fails to record their presence. Their cumulative dwell time remains below the required 40 minutes, and their attendance is automatically marked as **ABSENT / INVALID**.

2. COMPLETE BILL OF MATERIALS (BOM) & COST ESTIMATION

Below is the detailed itemized cost breakdown for building a single hardware node capable of monitoring a standard classroom room.

S.NO	COMPONENT NAME	KEY SPECIFICATION	QTY	EST. PRICE (INR ₹)
1	ESP32 Development Board	30-pin, Dual-Core, Integrated Wi-Fi + BLE 4.2 / 5.0	1	₹380 - ₹480
2	RC522 RFID Reader Module	13.56 MHz SPI Module + S50 White Card & Keyfob	1	₹180 - ₹240
3	0.96" OLED Display / 16x2 I2C LCD	128x64 I2C Blue OLED Display for status visual feedback	1	₹160 - ₹220
4	Status LEDs & Active Buzzer	Green/Red 5mm LEDs + 5V Active Buzzer for tap confirmation	1 Set	₹40 - ₹60

S.NO	COMPONENT NAME	KEY SPECIFICATION	QTY	EST. PRICE (INR ₹)
5	5V 2A DC Adapter / USB-C Cable	Power supply for uninterrupted room node operation	1	₹180 - ₹250
6	Breadboard / Prototype PCB & Wires	Jumper wires (F-F, M-F), soldering PCB, headers	1 Set	₹100 - ₹150
7	Custom 3D Printed Enclosure	PLA Wall-mount casing for entry door installation	1	₹200 - ₹350
TOTAL ESTIMATED COST PER ROOM NODE			--	₹1,240 – ₹1,750

* Note: RFID Tag cards cost approximately ₹15–20 per card for student distribution. If dedicated hardware BLE beacons are used instead of student smartphones/apps, add ~₹250–₹350 per BLE keyfob.

3. SYSTEM ARCHITECTURE & WORKING LOGIC

Hardware Layer

- MCU ESP32-WROOM-32U
- Entry MFRC522 SPI @ 13.56MHz
- Tracking BLE Scanning Engine
- Interface I2C OLED Display + Buzzer
- Network Wi-Fi 802.11 b/g/n (MQTT)

Software / Firmware Layer

- Logic Finite State Machine (FSM)
- Threshold RSSI ≥ -75 dBm
- Timer Cumulative 2400 sec (40 min)
- Sync MQTT / HTTP REST API to Server
- Storage EEPROM / Flash Local Backup

Finite State Machine (FSM) Transitions

- IDLE State:** System waits for a RFID tap. BLE scanner runs background sampling.
- CHECK_IN State:** Card ID detected & verified. Session created in ESP32 RAM with `dwell_timer = 0`. Visual feedback displayed.
- MONITORING State:** Periodic BLE scan checks if student's device/beacon $RSSI \geq$ threshold (-75 dBm). If detected, active time accumulates (+10s per interval). Short breaks (< 5 mins) pause accumulation without resetting.
- VERIFIED State:** Once total accumulated dwell time reaches **40 minutes (2,400s)**, attendance status updates to **PRESENT** and pushes record to cloud database.

4. IMPLEMENTATION HIGHLIGHTS & SCALABILITY

- Wall Bleed Mitigation:** Calibrating the BLE RSSI threshold (between -70 dBm and -78 dBm) ensures signals from adjacent hallways or neighboring classrooms are filtered out.
- Cost Efficiency:** By utilizing existing student smartphones as BLE advertisement beacons via a simple mobile app, additional hardware tag costs are eliminated.
- Cloud Integration:** Fully compatible with MQTT broker (Mosquitto/HiveMQ) and web dashboards (Node-RED, React, or PostgreSQL) for real-time institutional analytics.